

SYSC 5108 Exam Study: Assignment 2, Question 2

MLE for the Laplace rate parameter

Source question: SYSC 5108 W26 Assignment 2.pdf, Question 2

Question

Given n independent sample points $x_i, i = 1, \dots, n$, from a Laplace distribution with PDF

$$f(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \quad \lambda > 0,$$

provide the maximum likelihood estimator (MLE) for the parameter λ .

Step 1: Identify the model and parameter

The PDF is a zero-centered Laplace distribution:

$$f(x; \lambda) = \frac{\lambda}{2} e^{-\lambda|x|}.$$

The unknown parameter is λ , which controls how sharply the density is concentrated around zero. Larger λ gives a sharper peak and lighter tails.

Goodfellow, Bengio, and Courville define the Laplace distribution in Chapter 3 as

$$\text{Laplace}(x; \mu, \gamma) = \frac{1}{2\gamma} \exp\left(-\frac{|x - \mu|}{\gamma}\right).$$

The assignment's distribution is the special case

$$\mu = 0, \quad \gamma = \frac{1}{\lambda}.$$

So estimating λ is equivalent to estimating the inverse scale parameter.

Step 2: Write the likelihood

The samples are independent, so the joint likelihood is the product of the individual densities:

$$L(\lambda) = \prod_{i=1}^n f(x_i; \lambda).$$

Substitute the PDF:

$$L(\lambda) = \prod_{i=1}^n \left(\frac{\lambda}{2} e^{-\lambda|x_i|} \right).$$

Collect the repeated factors:

$$L(\lambda) = \left(\frac{\lambda}{2} \right)^n \exp\left(-\lambda \sum_{i=1}^n |x_i|\right).$$

Step 3: Take the log-likelihood

MLE usually maximizes the log-likelihood because the logarithm turns products into sums and does not change the maximizer. This is the same principle used in the Chapter 3 slides on MLE and in Goodfellow et al., Chapter 5, Section 5.5.

The log-likelihood is

$$\ell(\lambda) = \log L(\lambda) = n \log \lambda - n \log 2 - \lambda \sum_{i=1}^n |x_i|.$$

Step 4: Differentiate and solve the score equation

Differentiate with respect to λ :

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n |x_i|.$$

Set the derivative equal to zero:

$$\frac{n}{\lambda} - \sum_{i=1}^n |x_i| = 0.$$

Rearrange:

$$\frac{n}{\lambda} = \sum_{i=1}^n |x_i|.$$

Therefore

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n |x_i|}.$$

Equivalently,

$$\hat{\lambda}_{\text{MLE}} = \frac{1}{\frac{1}{n} \sum_{i=1}^n |x_i|}.$$

So the MLE is the inverse of the sample mean absolute value.

Step 5: Check that it is a maximum

The second derivative is

$$\frac{d^2\ell}{d\lambda^2} = -\frac{n}{\lambda^2}.$$

For $\lambda > 0$,

$$-\frac{n}{\lambda^2} < 0.$$

So the stationary point is a maximum.

One edge case: if every $x_i = 0$, then $\sum_i |x_i| = 0$, and the likelihood increases without bound as $\lambda \rightarrow \infty$. In that degenerate case there is no finite MLE. For ordinary data with at least one nonzero sample,

$$\sum_{i=1}^n |x_i| > 0,$$

and the finite MLE above is valid.

Step 6: Exam Interpretation

The negative log-likelihood is

$$-\ell(\lambda) = -n \log \lambda + n \log 2 + \lambda \sum_{i=1}^n |x_i|.$$

This shows why the Laplace distribution is tied to absolute error: the data-dependent term is proportional to

$$\sum_{i=1}^n |x_i|.$$

This contrasts with the Gaussian example from the Chapter 3 slides and Goodfellow et al., Chapter 5, where Gaussian maximum likelihood leads to squared-error terms.

A concise exam answer could be:

Since the samples are independent,

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda}{2} e^{-\lambda |x_i|} = \left(\frac{\lambda}{2}\right)^n e^{-\lambda \sum_i |x_i|}.$$

Thus

$$\ell(\lambda) = n \log \lambda - n \log 2 - \lambda \sum_i |x_i|.$$

Setting the score to zero gives

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_i |x_i| = 0,$$

so

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_i |x_i|}.$$

The second derivative is $-n/\lambda^2 < 0$, confirming a maximum.

Cross References Used

- Assignment: SYSC 5108 W26 Assignment 2.pdf, Question 2.
- Local submitted Assignment 2 PDF checked for comparison, Question 2 response.
- Slides: Chapter 3 slide deck, slide 26 on the MLE principle and maximizing the log-likelihood of independent samples.
- Slides: Chapter 3 slide deck, slide 27 on the Gaussian MLE example, useful as a contrast to the Laplace absolute-error form.

- Slides: Chapter 3 slide deck, slide 28 on MLE properties.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3, Section 3.9.4 on the Laplace distribution.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 5, Section 5.5 on maximum likelihood estimation and log-likelihood.